

GEN-AI POWERED DATA ANALYTICS

Presented by:

Syed Arshadul

25SCS1003005399

B.Tech CSE — IILM University



Introduction

The GenAI Powered Data Analytics Job Simulation by Tata iQ × Forage provided practical exposure to applying data analytics and Generative AI to a financial-services problem.

The simulation focused on using customer data to understand delinquency risk, generate actionable insights, and design a responsible AI-supported collections approach.

Focus: Data Analytics • Risk Prediction • GenAI • Responsible AI

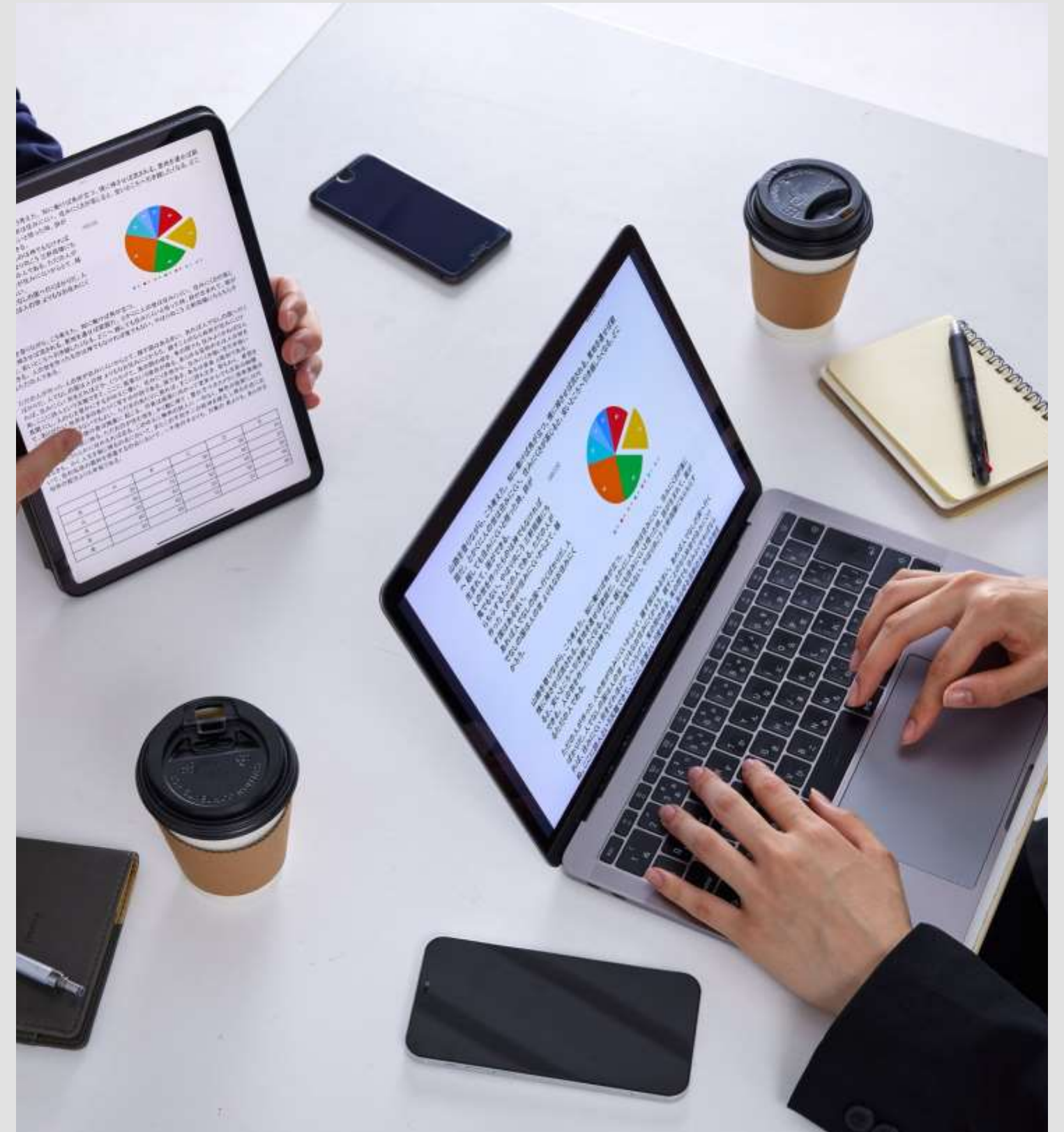


About the Internship

The simulation provided a practical business scenario involving Geldium Finance, where data analytics and GenAI were used to understand customer delinquency risk and develop a responsible AI-enabled collections approach.

Key Areas of Exposure

- Exploratory data analysis and data quality assessment
- Predictive analytics and delinquency-risk modelling
- Customer risk segmentation
- Business recommendations for collections
- GenAI-assisted analytical workflows
- Responsible and explainable AI implementation



PROBLEM STATEMENT

Identifying and Managing Delinquency Risk
Financial institutions need to identify customers who may be at risk of delinquency and prioritize appropriate collection strategies.

The challenge was to use data-driven insights and AI to support this process while maintaining fairness, transparency and human oversight.

Key Question:

How can analytics and GenAI help transform customer data into responsible, actionable collections decisions?



Objective

The primary objective was to develop a data-driven framework for identifying and managing customer delinquency risk.



01 Understand the Data

Assess data quality, completeness and important customer patterns.

02 Identify Risk

Explore predictive approaches for estimating probability and identifying risk .

03 Support Decisions

Translate analytical findings into customer segments and targeted collections strategies.

04 Responsible Automation

Design an AI-enabled workflow incorporating explainability, fairness, privacy

TASK/EXPERIENCE



Data Understanding & EDA

- Examined the structure and quality of customer data
- Identified patterns that could influence delinquency risk

⁰² Risk Prediction

- Identified potentially important risk-related features
- Focused on model interpretability and appropriate evaluation

Business Recommendations

- Converted analytical insights into actionable customer segments
- Identified relevant business performance

AI-Enabled Solution

- Incorporated GenAI as an analytical and decision-support layer
- Included human review for sensitive or uncertain cases

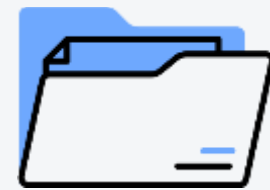
From Data to Decision



We addressed these challenges by combining data validation, interpretable analytical methods, business rules, human oversight and continuous monitoring.

Challenges Encountered

01 Data Quality



Real-world datasets may contain missing, inconsistent or incomplete information, which can affect analytical conclusions.

02 From Prediction to Action



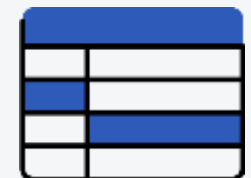
Identifying a high-risk customer is only the first step. The analysis must lead to an appropriate and practical intervention

03 Model Interpretability



A highly accurate model is not sufficient in a financial decision-making environment. Its predictions should also be understandable to stakeholders.

04 Responsible AI



Automating financial decisions introduces concerns around fairness, privacy, bias and accountability.



Responsible AI Framework

- 01 **Fairness**--Monitor whether decisions create unintended disparities
- 02 **Explainability**--Provide understandable reasons behind risk classifications
- 03 **Human Oversight**--Escalate sensitive and uncertain cases for human review
- 04 **Privacy**--Use appropriate data access and protection controls
- 05 **Monitoring**--Track model performance and changes over time
- 06 **Accountability**--Maintain auditable decision and model records

Experience & Learning



Technical Learning

- Exploratory data analysis
- Data quality assessment
- Predictive modelling concepts
- GenAI-assisted analytics



Professional Learning

- Structured problem solving
- Analytical communication
- Responsible AI thinking

Business Learning

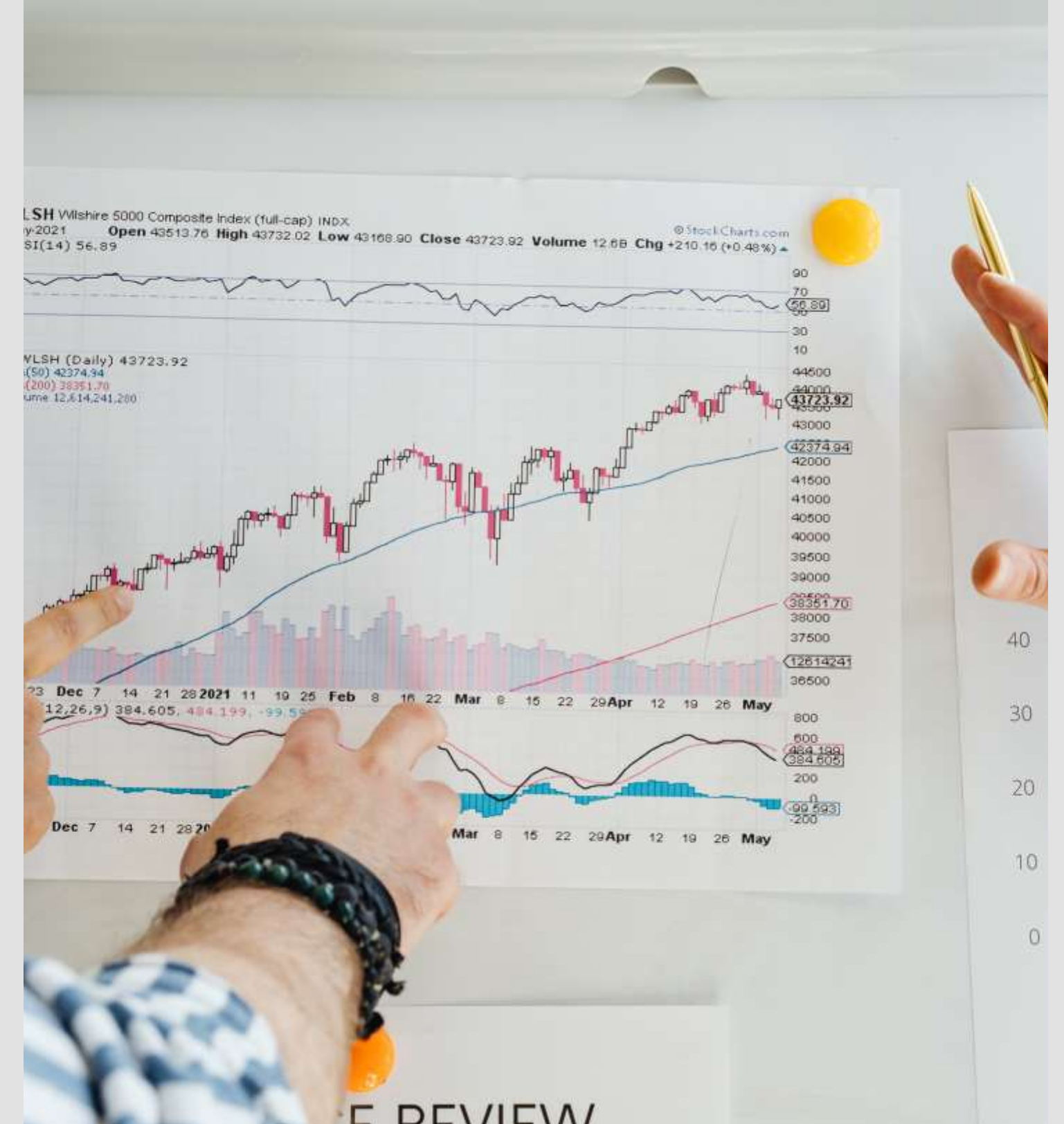
- Converting analytical findings into business recommendations
- Understanding stakeholder requirements
- Designing measurable interventions
- Thinking beyond model accuracy toward business



CONCLUSION

The simulation demonstrated how data analytics and GenAI can work together to solve a practical financial business problem.

It connected
data analysis → risk prediction → business
action → responsible AI
highlighting the importance of combining
technical capabilities with business
understanding and ethical decision-making.



Key Takeaway:

Effective AI is not just about prediction—it is about turning insights into responsible decisions.

Thank You

